



HRDWRE

CANDLEMAKER <3

Final Report Document

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Executive Summary

Purpose

The reader should understand what problem the venture is trying to solve and what Generate did to address this problem over the semester.

- Similar to a Lab Report's Abstract
- Aim for 250 words
- Communicate what problem the client is solving
- Communicate what steps Generate helped the product take
- This can be further research on the problem, prototyping to test a solution, creating foundations for the product, etc.
- Communicate how Generate planned to achieve that goal at a high level
- Touch on learning opportunities within the team

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Mechanical Design

Assembly Overview

Candle Maker consists of 7 subsystems: heating, dispensing, wick placement, mixing, lifting, cooling, and ejection. Through these 7 subsystems, wax pellets are heated up to around 180 degrees, dispensed into the wax chamber where a wick is placed and the scent and wax are mixed. The wax is then cooled and ejected from the wax chamber.

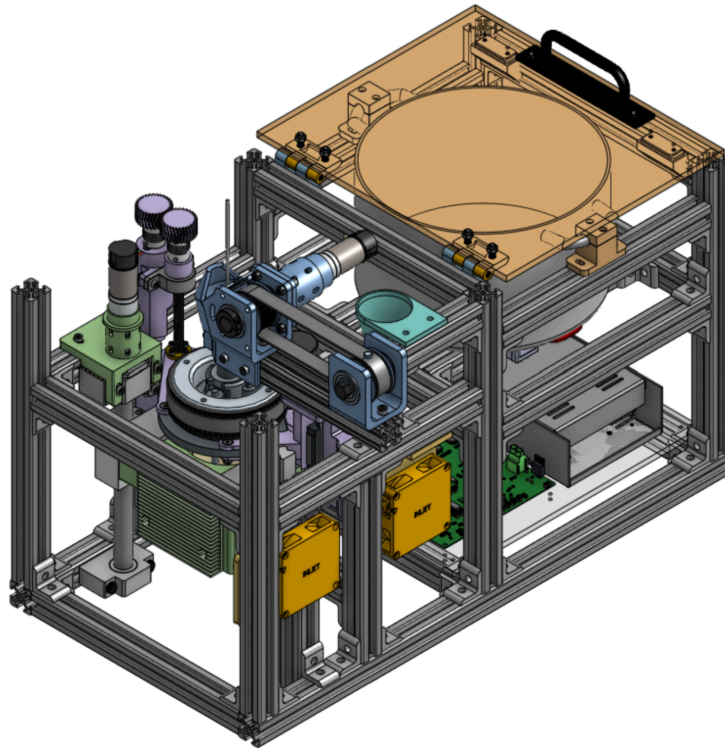
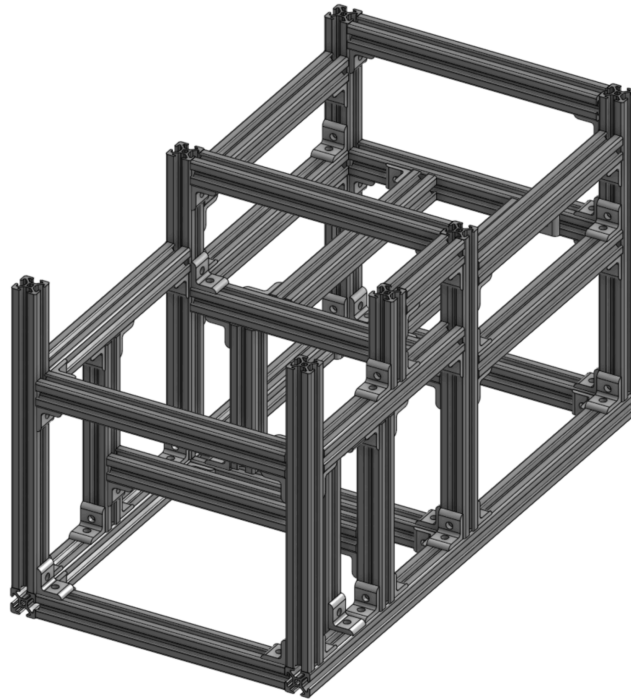


Figure 2: Candle Maker Top Assembly

Candle Maker was designed to be a small, table top appliance where a candle could be made at the touch of a button. Our approach to designing Candle Maker was to separate each part of the candle making process into its own subsystem. From this, certain aspects had to be condensed into one subsystem. All designs were made around one limiting factor, the size of the candle. With the goal of staying as small as possible, the wick placement and mixing subsystems were mounted to the lifting mechanism, which was placed above the wax chamber.

One shortcoming of this design is its use of 5 different motors, and complex assembly due to the small space used. Additionally, with the use of 3D prints and 8020, tolerances were disregarded from the start and ended up being a roadblock in the final assembly due to tight fits. With more time and thought out designs, more maneuverability could've been implemented, creating a smoother assembly process.

Structure



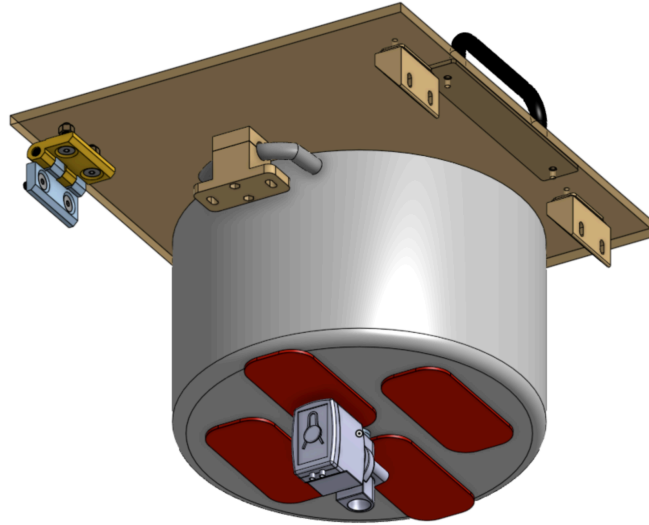
The structure of the Candle Maker was made out of 1 inch (1010) 8020 extrusions. 8020 is great for quick changes and allowing pieces to move within their slot that they use. While it can be hard to work with when assembling, it allows for easy changes of subsystems and gives wiggle room to tight tolerances. Additionally, all corner brackets and screws that would connect to 8020 were implemented in CAD so that we could see exactly where everything goes.

The entire structure was split into two cube-like areas. One was for the heating pot and all the electronics, and the other was for all subsystems related to making the candle. One goal at the start was to make sure that all electrical components had the space they needed to fit easily. Putting all these components together and under the heating chamber worked extremely well, allowing us to not have to worry about space constraints.

When assembling the structure, we found that splitting it up into sections worked well. This allowed us to have multiple engineers working on it at the same time. It also meant that if there needed to be changes, the entire structure didn't need to be taken apart to get to one small area.

Heating Assembly

■ Component details



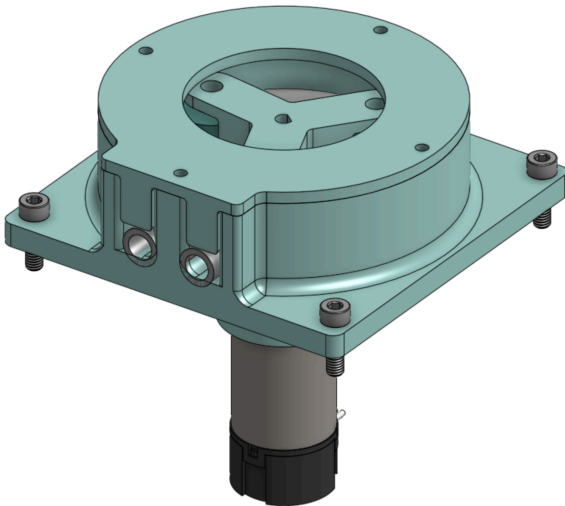
The heating assembly consists of a stainless steel pot that holds 5 quarts of wax and a valve at the bottom where the wax can enter the dispensing mechanism. Insulation material surrounds the sides and bottom of the pot while pot connectors attach it to the 8020 frame. The top of the pot is covered by a square top that attaches to the 8020 structure using hinges and secures closure with magnets. The user can use the handle to access the pot and pour in wax.

After considering off the shelf options of a heated and insulated pot with a spout, it was determined that it would be too hard to integrate into a tinier machine. Most off the shelf Presto Pots were too expensive or it would be hard to access their electrical components. After deciding to use an off the shelf cooking pot it was insulated, connected to the frame, had a spout, and had a pot top.

The stainless steel pot that was bought off the shelf is drilled and tapped so a brass spout with threads can connect on the bottom. A solenoid is connected to the bottom of the brass spout. Pot connectors connect the handles to the 8020 framing.

An underestimation of the pot handles in CAD led to the overall frame structure to be smaller than it should've been. To resolve this problem, the pot was turned diagonal and the connectors were designed with one wing that angled 45 degrees. The magnet holders and hinges were moved closer to each other so that the connectors would fit in.

Dispensing Assembly



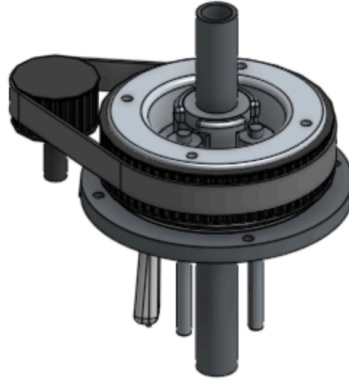
The primary function of the peristaltic pump is to supply consistent, controlled volumes of molten wax to the mold station for the entire candle-making machine. It functions as a critical interface between the melting reservoir of wax and the mixing operation in the system. Using a rotating compression system, the pump transfers the melted wax with its high viscosity through flexible tubes without directly contacting any mechanical component, maintaining purity and reducing the possibility of clogging.

It is positioned in between the melting chamber and mixing assembly on the 8020 frame.

What is this design's function? Relate to the overall assembly

- How did you get here in the design process?
- Key features of design
 - Ex. Material Selection--why is this the best choice? Cost, performance, etc.
 - Ex. Tolerances

Mixing Assembly



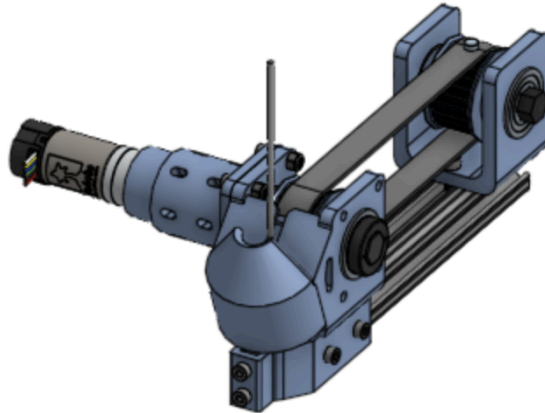
The mixing subsystem is responsible for the agitation and combination of both the melted wax and custom essential oils within the wax chamber. The mixing assembly is mounted to the lift assembly, allowing it to be elevated into and out of the wax chamber during the mixing and cooling processes. Additionally, the mixing chamber is kept from accumulating wax through a compliant scraper which circulates around the edges during mixing.

After careful consideration of the many motorized possibilities to mix the oil with the wax, such as a differential wrist and position offset motors, a planetary gearbox drive was selected for use in the final design. Wick placement's necessity of an unobstructed concentric axis to properly align the wick in the center of the candle could be integrated into the planetary system without obstructing the mixing process itself. The planetary gearbox accomplished this through the sun gear being fixed upon a hollow aluminum shaft, with the ring gear being driven through a PLA printed pulley atop an OTS lazy susan bearing. The system was lubricated fully internally. With a gear ratio of 48:26, each of the three PLA printed planets were press fit onto an aluminum D-profile rotary shaft which would interface with the wax. The entire system pulled support from the lazy susan bearing, which was mounted to the lift assembly via four through holes. The entire assembly required tight tolerances and testing to reduce friction and interference between moving elements. In a future iteration, better attention should be paid to designing for assembly, as the tight tolerances required impractical assembly methods. Additionally, a better methodology for gear alignment should be developed, along with better material selection for them which will improve longevity.

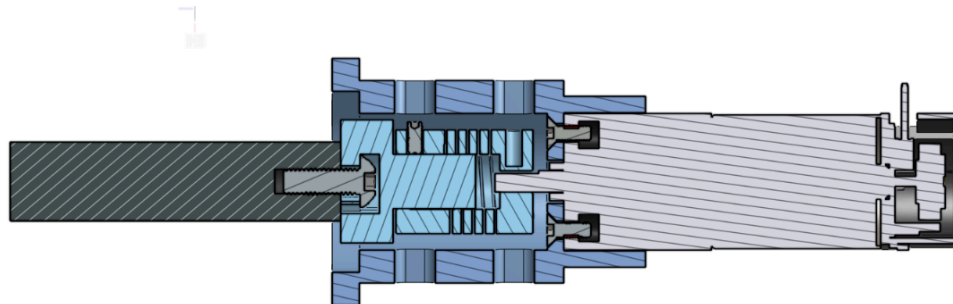
Attached to the bottom of the mixing assembly was the compliant scraper, which was constructed of a silicon-overmolded stainless 304 sheet metal bent part which was outsourced for manufacturing. With the metal part fabricated, the compliant portion was modeled and used as a negative when modeling the mold structure. The scraper is mounted through the same

hardware which holds together the planetary system.

Wick Placement Assembly

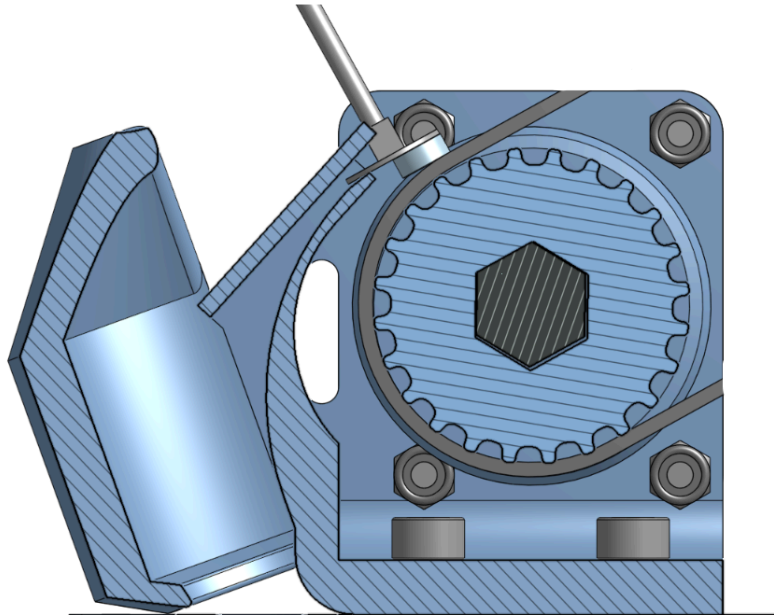


The wick placement assembly had the goal of holding multiple wicks, and inserting one into the center of the wax chamber each time a candle was about to be made. This assembly consists of a motor and motor coupler, 3D printed structures, two pulleys, and a belt with magnets on it. The motor rotates the belt and the wicks that are placed on magnets get picked off by the 3D printed ramp where the wick then falls down the mixing assembly's center shaft, into the wax chamber. To keep the belt in tension, a piece of 8020 was used to connect the two pulley assemblies so that one of the pulleys could be slid farther away and therefore tension the belt.



Seen above is the coupling from the motor to the hex shaft that rotates the pulley that connects to the belt with the glued magnets. Due to the size of the hex shaft and the shaft couplers that we bought, an intermediary piece was created to allow us to convert the rotation of the motor into the hex shaft. To get to the set screws in the shaft coupler, slots were made in the 3D printed housing that encases it. One problem with this design was the stack up of different components and their tolerances. If they weren't perfectly aligned, the motor would have to do more work and would often stall out.

The way the wick would be taken off the magnet and fall into place was entirely designed around the wicks dimensions. Using a magnet that is smaller in diameter than the wicks metal base, it was able to be “clipped” and fall down a ramp into the mixing tube. This idea can be seen in the picture below.

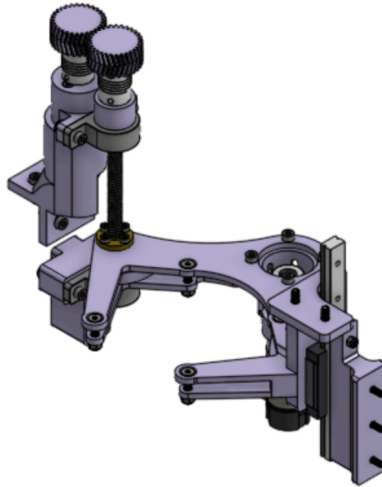


This was only possible because we used a wick with a magnetic base. One downfall of doing this was that the wick was very top heavy, so it wouldn't get fully down the ramp before falling over and getting itself stuck. There was a thought of placing magnets in other areas to keep it from tipping but there was not enough time to implement this.

There were many different design ideas that were thought of revolving around rotational sorting, or a vending machine screw design, but the design chosen was the simplest and most effective. It had a small parts list consisting of mostly 3D printed parts. This made it easy to iterate quickly.

Lift Assembly

The lifting mechanism's goal was to combine the vertical movement of mixing and wick placement into one. Taking inspiration from the ejection subsystem, the lift assembly consists of one lead screw and one linear rail to do this vertical movement. The motor is connected to the lead screw and the linear rail is used to keep the whole system level. This system is the intermediary part of the entire candle making process. It is the piece in the middle that brings everything together.

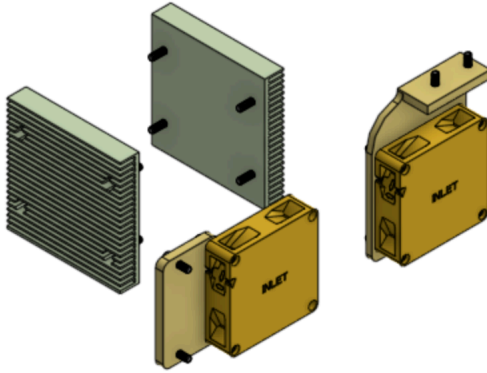


The assembly can be divided into several components consisting of the lift motor mount, herringbone gear drive, lead screw, mounting plate, mixing motor mount, and the linear slide mount. The mounting plate is the focal point of the assembly, providing an interface between the mixing subsystem's lazy susan bearing and the lift itself. The plate allows for the lazy susan bearing to slide into the gap created by the space between the two plate segments, secured via four screw hole locations. The mixing motor mount is located on the backside of the mounting plate with the motor being affixed from beneath. This orientation allowed for the mixing drive pulley to be properly aligned with the driven pulley given space constraints. The herringbone gear drive, receiving power from the associated motor mount, powers the lead screw which is used to move the entire assembly up and down. This combination provided a small form factor while also utilizing herringbone gears to guarantee proper interfacing. Lastly, the linear slide and mount provide stability to the opposite side of the lead screw while allowing a smooth linear motion during the raising and lowering processes. Utilizing the 8020 structure, the lead screw and motor are mounted on one side, and the linear rail is mounted on the other.

In a future iteration, the team would have liked to place more emphasis on tolerancing. The many moving and interfacing parts meant that any misalignment or part dimensions could cause major problems for the entire assembly. Some minor tweaks such as an improved lift motor mount and creating slots for proper mixing pulley tensioning would drastically improve the functionality of the lift.

Cooling Assembly

- Component details



- What is this design's function? Relate to the overall assembly

Cooling is used to cool the candle faster and shrink the candle a little bit so it pops out of the chamber. Fans mounted to the 8020 framing blow across heat sinks that are attached to the chamber. Thermal paste conducts the heat from the chamber to the heat sink.

- How did you get here in the design process?

The choice for cooling was originally between the two PC cooling methods, water cooling and fan cooling. After testing different methods of cooling candles it was found that using fans was effective in cooling the candle without making it crack. The fan cooling would also be easier to integrate.

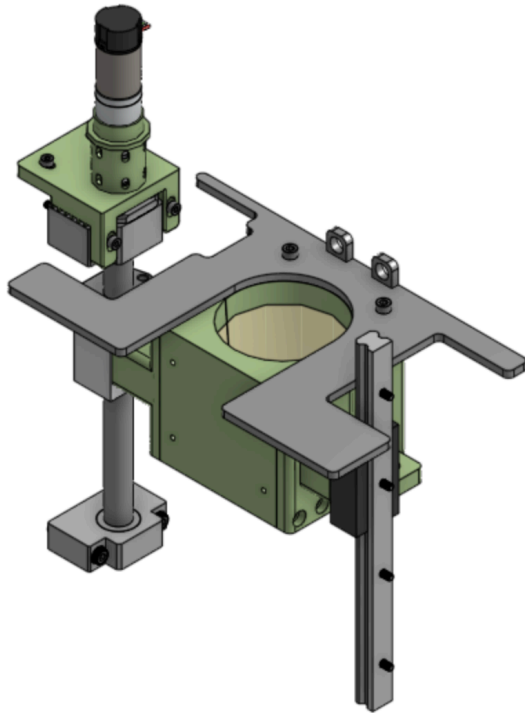
- Key features of design

- Ex. Material Selection--why is this the best choice? Cost, performance, etc.
- Ex. Tolerances

A 90mmx90mm heat sink is chosen so that the heat sink is the size of the chamber exactly. The heat sinks were drilled in from the back and a socket head cap screw with a washer attaches the heat sink to the chamber. The fans mounts were made considering the fan's width. The fan's don't stick out so that the acrylic can be flush against the structure.

Max Chamber Assembly

- Component details



bvfgc This subsystem supports and holds the candle while it is being made. The chamber-like cylinder holds the melted wax being pumped into it, supports the wick during placement, and contains the mixing section which is important to the candle-making process. Beyond serving as a mold during formation, the design also enables ejection once the wax has cooled using heat sinks. The ejection mechanism features a vertical linear rail, allowing the chamber to split into two parts. Once cooled, one part displaces vertically from the other, enabling the user to easily and efficiently retrieve the finished candle.

Several design iterations were explored before finalizing the current system. Early concepts included a horizontal linear rail, but that created uncertainty regarding which side of the chamber the candle would stick to upon cooling. Another approach involved pushing the candle out using a linear actuator so it could fall onto a surface for user collection but this method unnecessarily complicated and made user retrieval more difficult. Before settling on linear rails, mechanical solutions such as worm gears and rack and pinion systems were considered to convert rotational motion into linear displacement. While those were technically feasible, a combination of a linear rail and lead screw for vertical motion was more effective. This approach uses gravity to assist with separation and ejection, simplifying the mechanism while maintaining functionality.



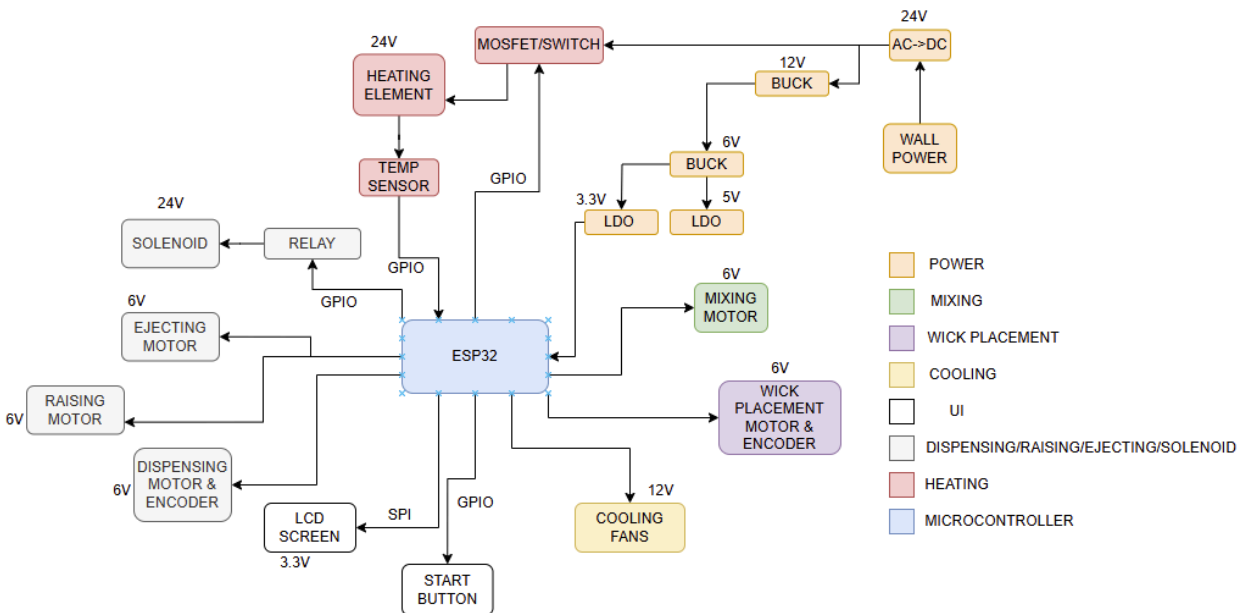
The finalized design incorporates several key components. The main chamber is split into two parts joined at a 5-degree angle, allowing the halves to stick together through friction alone. To center the wick during placement, a built-in magnet is added at the bottom of the chamber. Additionally, a silicone layer is placed between the two parts to provide extra grip and sealing. The silicon was so effective that even water cannot pass through without extra force pushing the two pieces together. The stationary portion of the chamber also includes a feature for attachment to an 8020 beam.

Connectors are important for attaching the chamber to the linear rail and lead screw. Two unique designs serve this purpose: one links the chamber to the rail block, and the other connects it to the lead screw block. A mechanical stop is also included to prevent users from touching other subsystems. This feature ensures user safety by limiting interaction to the candle itself and also includes supports for tubing that carries melted wax and scent. Finally, the motor holder secures the lead screw motor that drives vertical movement. It holds the motor, provides room for a coupler that connects the motor to the lead screw, and includes elements compatible with 8020 beam integration.

The design balances cost, performance, and tolerance by using a mix of SLA and FDM 3D printing. High temp SLA is used for the main chamber to handle heat without warping, while other parts are printed with more affordable materials. Important tolerances include the chamber split and silicone seal which need to be tight enough to prevent leaks but still allow easy disassembly. Precision is also maintained at motor and rail interfaces to ensure precise movement of the chamber when it opens and closes.

Electrical Design

Design Overview



Candle Maker Electrical System Block Diagram

The block diagram above illustrates the high-level architecture of the candle-making prototype system. At the core of the system is the **ESP32 WROOM-32E** microcontroller, which orchestrates control and communication between all hardware subsystems. The ESP32 communicates with motors, sensors, a display, a start button, and other peripherals via GPIO, SPI, and PWM interfaces. Power is delivered through a 24V AC-DC converter, and then regulated to appropriate voltages using a combination of buck converters and low-dropout regulators (LDOs).

Subsystems are categorized based on function—heating, dispensing, mixing, wick placement, cooling, and user interface—and are each powered and controlled independently, though coordinated by the state machine running on the ESP32.

Design Objectives and Component Selection

The primary design constraint for the prototype was **simplicity** to ensure rapid development and increase the likelihood of having a working system within the semester timeframe. Accordingly, all hardware and software decisions were driven by ease of integration, affordability, and proven reliability in prototyping contexts.

- **Microcontroller:** The ESP32 WROOM-32E was selected for its extensive GPIO support, dual-core performance, built-in Wi-Fi and Bluetooth capabilities, and compatibility with both the Arduino IDE and PlatformIO. Its support for FreeRTOS and PWM control made it ideal for multi-threaded motor and sensor control tasks.
- **Heating System:** Off-the-shelf 24V heating pads were used due to their cost-effectiveness and compatibility with basic switching circuits. A high-current MOSFET is used to switch power to the heating element, and feedback is provided by a thermistor connected to an analog input. This approach avoids the need for expensive heating cartridges or specialized controllers.
- **User Interface:** The Waveshare 2-inch LCD screen was chosen for its low cost and ease of development using SPI communication. The TFT_eSPI library simplifies graphical output and menu handling, enabling responsive and informative feedback during system operation.
- **Power Distribution:** A 24V wall-powered AC-DC adapter supplies the system's primary voltage. This voltage is then stepped down using dedicated buck converters to produce 12V and 6V rails for motors and fans. Two 5V and 3.3V LDOs provided power to components, sensors, and the display. Using modular converters eliminated the need for designing custom power management ICs and simplified system integration.
- **Motors and Encoders:** Five 6V brushed DC motors from Pololu were selected for motion-related tasks, including dispensing wax, mixing, raising/lowering mechanisms, and ejecting candles. Four of the motors include quadrature encoders for position feedback, enabling more precise control. By standardizing on a single motor type and supply voltage, motor control and driver design were significantly simplified.
- **Solenoid and Relay:** A 24V solenoid valve is controlled via a relay circuit triggered by a GPIO pin. This isolates high-current switching from the microcontroller and allows safe and reliable actuation.
- **Expandability:** Due to limited GPIO availability on the ESP32, two GPIO expanders were incorporated to interface with additional sensors, limit switches, and peripherals.

Electrical I/O and Subsystem Integration

The system's hardware and software are tightly coupled through defined I/O channels:



- **Heating:** Controlled via a PWM signal to the MOSFET switch. Temperature readings are processed from the thermistor through analog input.
- **Motors and Encoders:** Motors are driven via H-bridge driver ICs controlled by GPIOs. Encoders return pulse signals to interrupt-capable pins on the ESP32.
- **Solenoid Valve:** Actuated through a relay controlled by a digital output pin.
- **Cooling Fans:** Powered directly by the 12V rail and switched via a GPIO-controlled transistor or MOSFET.
- **LCD Screen:** Communicates via SPI and is updated based on the current state of the system.
- **Start Button and Limit Switches:** Connected to digital inputs with internal pull-ups and polled or interrupt-driven in software.

Each component is designed to map logically into the overarching state machine implemented in software, where state transitions dictate the activation or deactivation of motors, heating elements, and other hardware subsystems. This structure ensures modularity, making the system easy to debug, expand, and refine in future iterations.

- Flowchart of how components and subsystems interact
- Explanation of decisions made in using general technologies and concepts
- I/O of the electrical systems in relation to mechanical, software, etc.

Power

Flow Chart (replace this with a flow chart)

Purpose

The subsystems of Candle Maker require different voltage levels for power. The buck converters implemented are required to change 24 V from the power supply to the voltage levels different parts require. The 3.3V buck converter provides power to logic level components,

Placement

The power network is placed in close proximity to the PCB power supply input. This minimizes thermal or power losses on the highest voltage level net.

Motors



172:1 Metal Gearmotor 25Dx71L mm HP 6V with 48 CPR Encoder

Purpose

The Candle Maker incorporates 5 brushed Pololu gearmotors to drive each of the mechanical subsystems. Each motor runs using 6 volts and may pull up to 6 amps (while stalling). The raising and ejection subsystems use a gearmotor with a 171.79:1 gear ratio, resulting in a no load speed of 57 RPM and a stall torque of 27 kg*cm. The remaining subsystems, mixing, dispensing, and wick placement, have a smaller torque requirement, so they use a 46.85:1 gearbox, resulting in a 210 RPM no load speed, and a stall torque of 9.1 kg*cm.

Every motor except the mixing motor also includes an attached encoder for accurate positional control, with up to 48 steps per rotation ungeared. With the gear box, the number of steps per rotation is $48 \times \text{the gear ratio}$ ($48 \times 171.79 = 8245.81$ steps/rev for the 172:1 gearmotor).

Placement

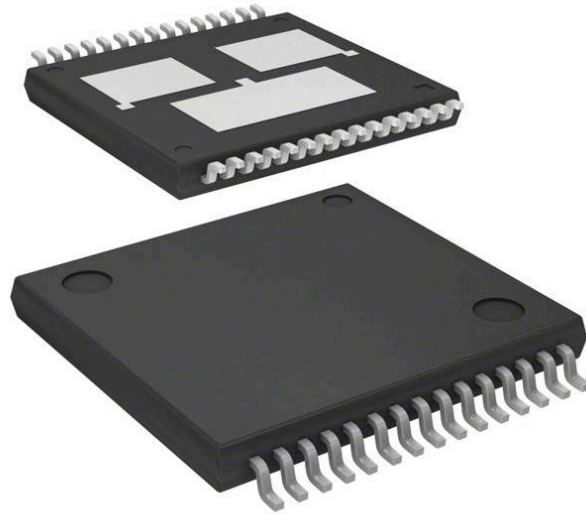
Each motor is placed within the mechanical subsystem that it drives.

Spec Sheet and Notes

The data sheet for the motors can be found below:

<https://www.pololu.com/file/0J1829/pololu-25d-metal-gearmotors.pdf>

Motor Driver



VNH5019ATR-E Motor Driver

Purpose

Each of the 5 motors are controlled by its own VNH5019ATR-E Motor Driver (for a total of 5 motors drivers). The Motor Driver used is a fully integrated H-bridge driver that controls the motors through PWM. The use of the driver allows for speed and directional control of the motors. The driver works with 2.5 - 5V logic so it is powered using the 3.3V for the microcontroller. The driver works with 5.5-24V motors and up to 12 amps.

Placement

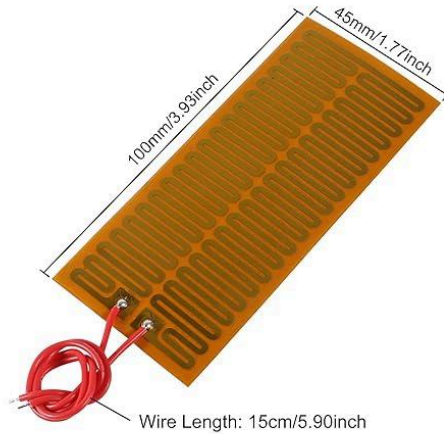
5 VNH5019ATR-E Motor Drivers (for each of the corresponding subsystems) are placed on the PCB.

Spec Sheet and Notes

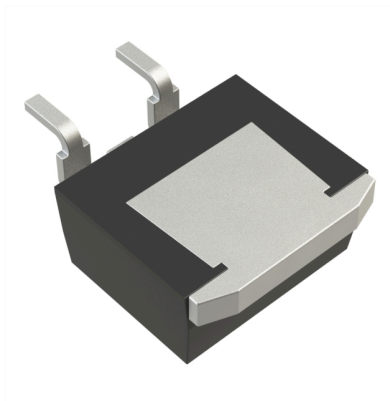
The data sheet for the VNH5019ATR-E Driver can be found below:

<https://www.st.com/content/ccc/resource/technical/document/datasheet/1c/26/60/a8/3e/70/4c/e1/CD00234623.pdf/files/CD00234623.pdf/jcr:content/translations/en.CD00234623.pdf>

Heaters



Film Heater



SMD FDB3632 MOSFET

Purpose

The heaters provide the electrical components that can melt candle wax for efficient deposition, scent infusion, and wick placement processes. The FDB3632 MOSFET was used for its relatively low threshold voltage, low susceptibility to high temperatures, and high switching current. This allowed the heaters to be driven with their maximum power, allowing rapid and efficient melting of the wax.

Placement

Because flexible heating pads with adhesive were chosen, they were attached to the sides and bottom of the wax containment chamber. The bottom in particular was chosen to be able to liquify the wax before depositing it into the forming chamber using the pump.

Spec Sheet and Notes

The first heating element considered was NiCr wire. NiCr wire is commonly used in industrial heating applications due to its simplicity of use, cost-effectiveness, and easy replaceability. These were replaced with film heating elements for two reasons: electrical compatibility and adhesive compatibility. When the NiCr wires were initially tested, it was discovered that an electrically insulating, thermally conductive adhesive would be needed to affix the NiCr wire to the pot. The film heating elements solved this and provided a more numerically stable power output to use in calculations, with each film element able to deliver 30W of power if driven at 1.25A DC and 24V.

Temp Sensors



NTCLE213E3303HHB0 Thermistor

Purpose

The temperature sensors were included in the heating subsystem to provide accurate and instantaneous monitoring of the temperature of the wax within the wax containment chamber.

Placement

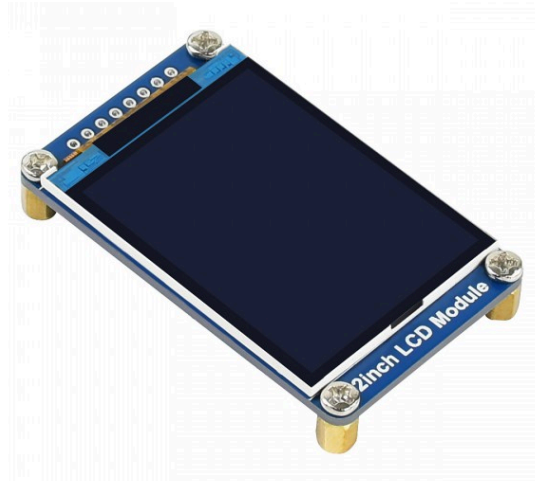
Placement of the temperature sensors is an ongoing process due to the highly variable nature of the temperature across the pot, low thermal conductivity of air, and sensitivity of temperature sensors to coating with wax. For this iteration of the system, the placement was chosen such that the bead of the temperature sensor was touching the pot.

Spec Sheet and Notes

A negative temperature coefficient thermistor was chosen as the type of temperature sensor for the project due to its efficient cost, simple integration, easy replaceability, and low power draw. The NTC thermistor functions as a variable resistor, where resistance decreases exponentially with temperature. Other options for temperature sensors include the relative of NTC

thermistors, PTC thermistors, resistor temperature sensors (RTD), thermocouples, and IR temperature sensors.

LCD Screen



Waveshare's 2-inch LCD Module

As touched on above, the 2-inch LCD module from Waveshare was selected to display the real-time states of the Candle Maker, including stages such as idle state, heating, wick placement, dispensing, mixing, cooling, ejecting, and the complete state. The display was programmed using the TFT_eSPI and LVGL libraries, creating graphical elements and text, providing the user with feedback on the machine's operation.

The LCD screen was positioned to be mounted on the front of the Candle Maker to ensure optimal visibility for the user.



Original Design of UI

The initial plan for the design of the user interface was to utilize the LVGL Image Converter, a free tool provided by the LVGL ecosystem, which can transform images into C array representations which can then be used with LVGL's image object, allowing you to easily directly display the image on the screen. Shown above, designs were created to showcase the individual states of the machine. However, it became apparent that the ESP32's limited flash memory posed a significant constraint as each converted image occupied a large chunk of space. With more than a few images included, the ESP32's available flash capacity was exceeded and resulted in failed builds. Due to memory limitations, the approach was replaced. Instead of relying on pre-rendered images, the interface was restructured using LVGL's built-in object system, which required significantly less flash usage. Each screen state was dynamically rendered with *lv_label* for text, *lv_bar* to create progress indicators, and *lv_obj* components for layout and custom visual elements.

Fans



12V DC Fan from Delta Electronics

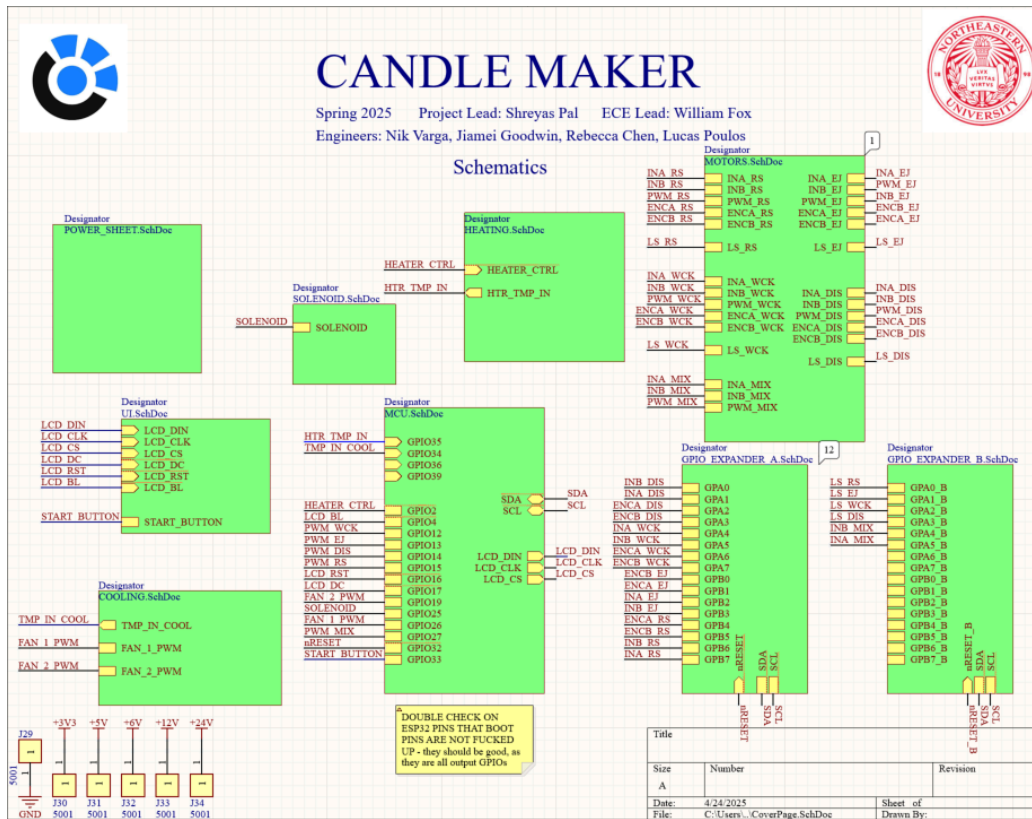
For the cooling subsystem, 12V DC fans from Delta Electronics were selected for their cost-effectiveness, reliability, and ease of control. These fans were particularly suitable for the system because their speed could be controlled by the ESP32 via PWM (Pulse Width Modulation). This allows for the speed of the fans to be dynamically adjusted. The fans were mounted parallel to the heat sinks so that wind could be blown across to the chamber holding the candle for passive and active cooling.

Initially, a water cooling system was considered. However, this approach raised concerns about the possibility of overcooling the wax. Cooling wax too quickly is known to cause issues such as cracking, uneven setting, sinkholes, shrinkage, frosting, and other issues concerning the structural integrity of the candles.

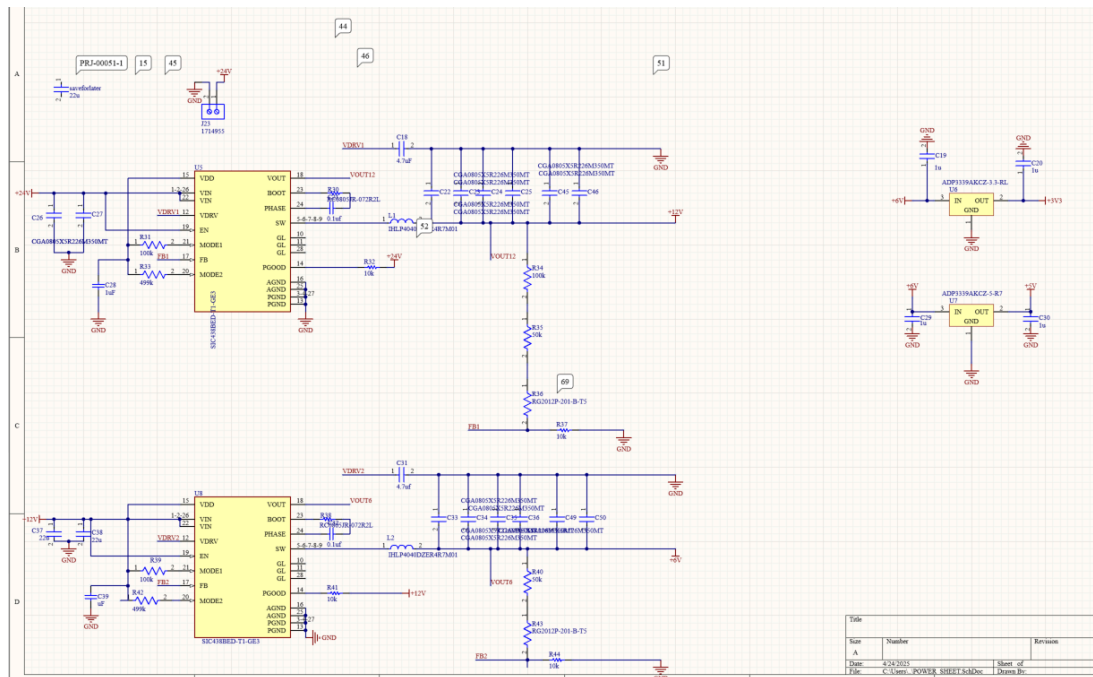
Circuit Board

Throw the main page of the schematics and then all the sub pages and a pic of the board

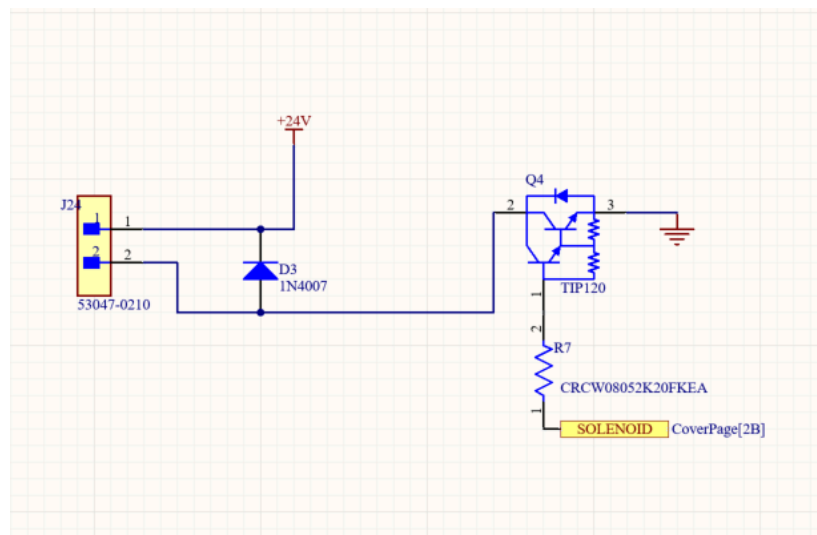
The following images depict the schematics that define the board as well as the board itself.



Schematic Cover Page

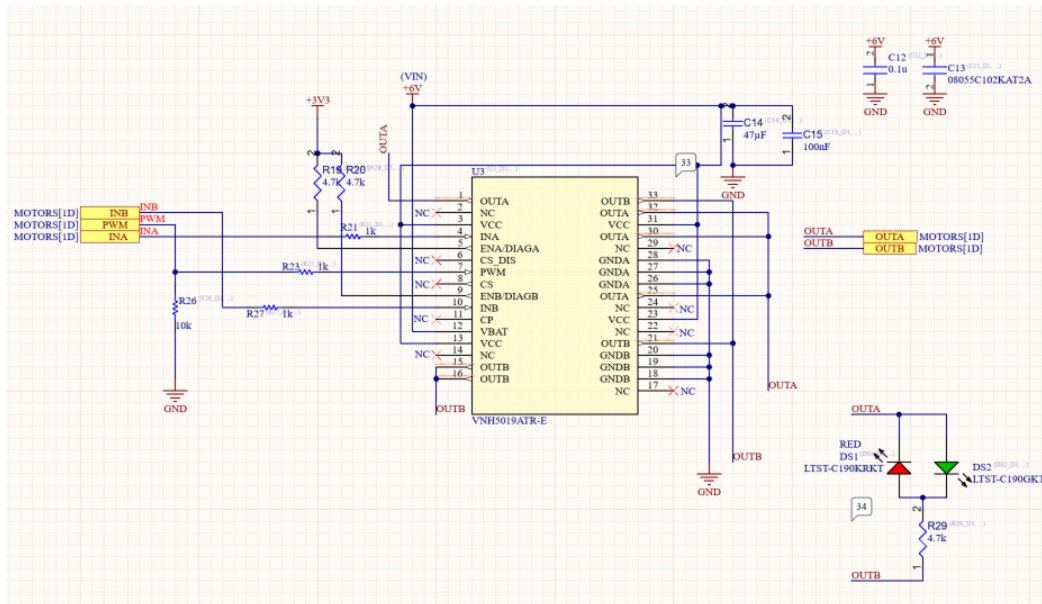


Power Schematic

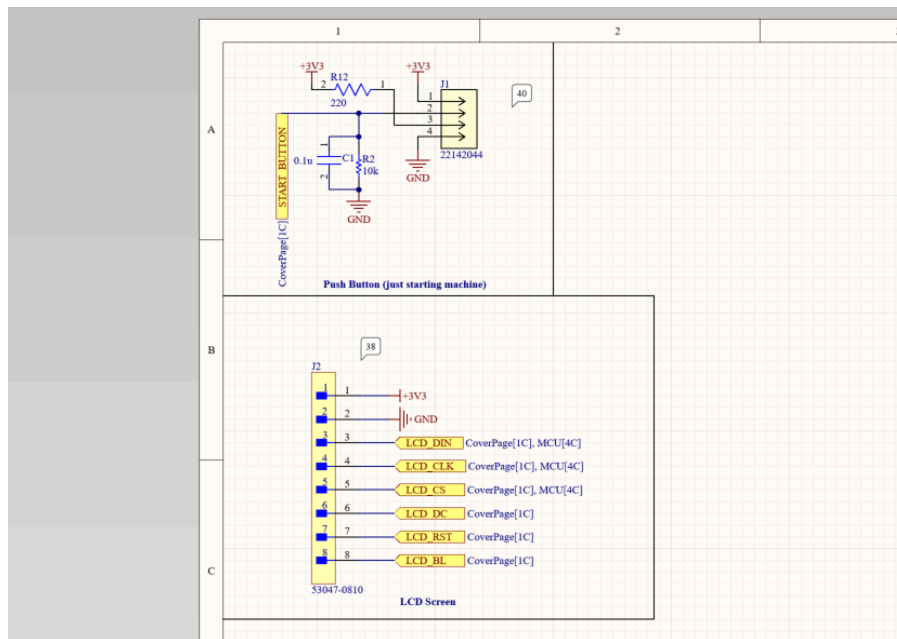


Solenoid Schematic

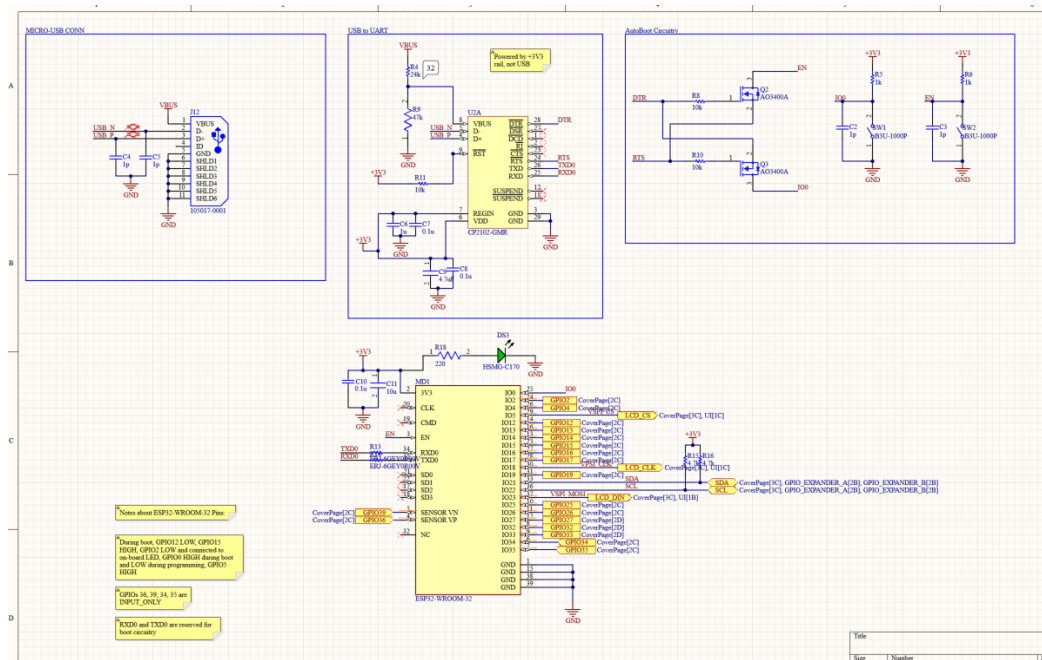




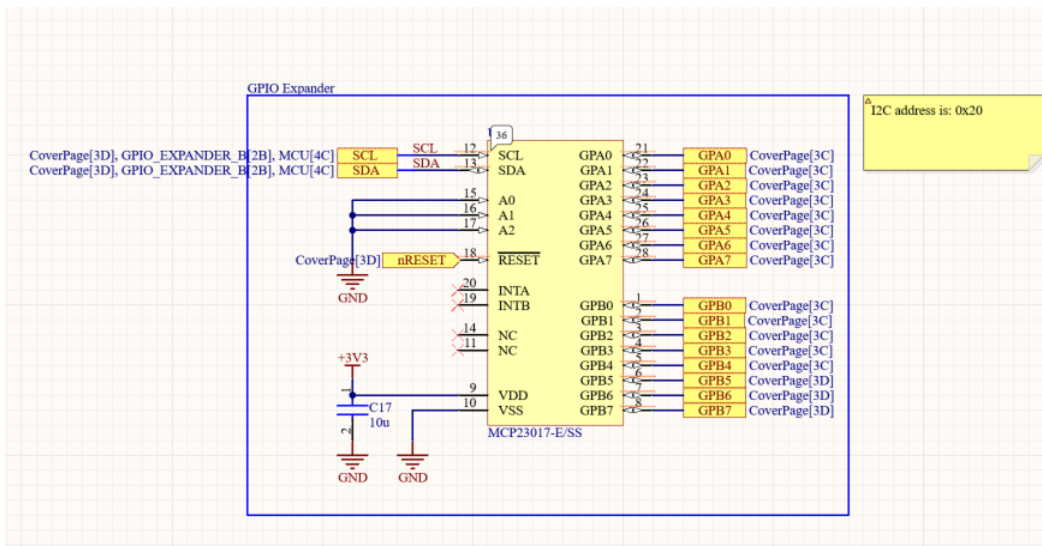
VNH5019 Motor Driver Schematic



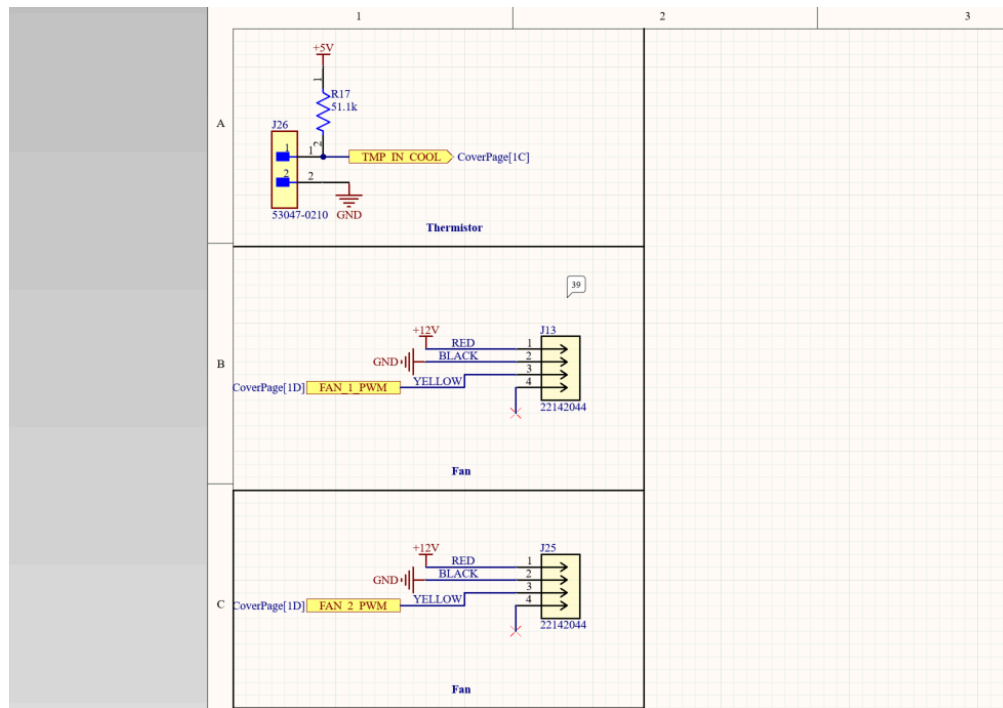
User Interface Schematic



Microcontroller Schematic



GPIO Expander Schematic



Cooling Subsystem Schematic



Software Design

General Software Design

PID Class:

The `PID` class is designed to implement heating functions, which are controlled through PID (proportional, integral, derivative) control. The PID code utilizes the z-transform of the continuous time PID equations derived in this Phil's Lab video. Critical functions are `readThermTemp()` and `update()`. The `readThermTemp()` function uses the voltage divider equation, hardcoded resistor values, hardcoded V+ values, and the coefficients characteristic to the NTCLE213E3303HHB0 thermistor to determine the temperature it is measuring. The `update()` function calls this to determine the error between the measured temperature and the setpoint and generates an output using this. The empirically determined limits and coefficients for the proportional, integral, and derivative terms are set in `CMStateMachine.cpp`. The `update()` function is configured in a freeRTOS task which is called based on the variable `samplingInterval` such that every `samplingInterval` milliseconds, the PID error and output are updated. This allows for a constant sampling rate, which makes the PID system time invariant and causes it to work. The task also allows the PID to operate outside of the structure of the main loop, which means it can run in parallel with other tasks the state machine needs to perform. The code for this is implemented in `pid.hpp`, `pid.cpp`, and `CMStateMachine.cpp`.

LCD Class?:

Encoder Class? :

Motor Class?:

Solenoid Class:

The `Solenoid` class is a lightweight wrapper designed to control a digital output pin connected to a solenoid valve. It abstracts the low-level pin control into two simple methods: `activate()` and `deactivate()`, which set the pin `HIGH` or `LOW` respectively. Upon instantiation, the class stores the assigned GPIO pin and initializes it as an output. This modular design allows the solenoid to be easily integrated into different states of the state machine where valve control is needed, such as during the `DISPENSING` stage.

fox.wi@husky.neu.edu what else am i missing?



Explanation of Overall Software Decisions

The overall software design consists of a central state machine that governs the behavior of the system, with each peripheral device (such as the motor driver, encoders, and screen) managed through its own dedicated class or module. This modular structure allows for cleaner abstraction and separation of responsibilities, making the system easier to scale and debug.

The state machine transitions through a predefined sequence of operational states: **STANDBY**, **HEATING**, **DISPENSING**, **MIXING**, **WICK_PLACEMENT**, **COOLING**, and **EJECTING**. Each state is responsible for orchestrating the behavior of specific hardware components necessary for that part of the process. For example, in the **HEATING** state, a PID controller is used to maintain a target temperature by modulating a heating element via PWM. Once each state completes its task or reaches a condition, it transitions to the next one based on a pre-set transition map.

Communication & Command Handling

The system is also designed to support external communication, potentially over a serial interface (e.g., USB or UART), to allow external control, diagnostics, or real-time monitoring. A simple command set is being designed to allow interaction with the machine during development and debugging.

Command Set (Planned)

- **START**: Begin the process from **STANDBY** to **HEATING**.
- **STOP**: Immediately halt the process and return to **STANDBY**.
- **STATUS**: Return the current state and any key sensor readings (e.g., temperature).
- **SET TEMP <value>**: Set the desired target temperature for the PID loop.
- **DEBUG PID ON/OFF**: Enable or disable live PID tuning/debugging output.

How Commands Are Interpreted

Incoming commands are expected to be parsed via the serial port. The command handler breaks down the string into tokens and matches the first token to a known command. Arguments are



validated and passed to the appropriate module or state logic. For safety and reliability, only certain commands are accepted in certain states—for instance, **SET TEMP** will only be accepted in **STANDBY** or **HEATING**, and not during **DISPENSING** or **MIXING**.

Although this system is still under development, the intended architecture provides a structured and maintainable way to control a complex, multi-step process with clearly defined state transitions, hardware abstractions, and external command interfacing.

SPI Class:

SPI communication was used to send data from the ESP32 to the Waveshare 2 inch LCD screen, which uses the ST7789VW chip to map the incoming data to pixels on the screen. Conveniently, the library “Bodmer/TFT_eSPI”, which is available via the Arduino IDE or Platform.io in VSCode, handles all communication for the screen and a SPI class is therefore not needed. However, since this library is licensed under GPL-3.0, if it is used in the final source code, the full source must be made publicly available under the same license; otherwise, a custom SPI communication class must be implemented to avoid the license requirements.

Testing/Version Control:

Throughout the process of software development, features were iterated using Git version control and branching. This allowed precise control over feature development, gradual integration of features as they came online, and a development process that was well integrated with testing and debugging the electrical system. Branches were used for developing different subsystems, such as UI and motors, but branches were also used to test features on the hardware. As features were tested successfully, they were integrated into the main branch.

Next Steps

- Define exactly where the project stands
 - What works? What doesn't?
 - What are the existing projects / shortcomings?
- What do you recommend doing next?
 - Getting quotes from manufacturers?
 - Gathering user feedback
 - What would you do if you had more time to work on the project?

Heating:

Future iterations of the heating system should consider more locationally targeted heating elements to deliver uniform heating across the wax containment chamber. Care should be taken to mechanically insulate the heating system to distribute heat uniformly. Placement of the thermistor is another critical area of improvement for the project. The sensor needs to be protected from environmental extremes such as temperature and wax coating, but also needs to measure the temperature of the wax accurately.

Software:

On the software side, improvements need to be made to the codebase to enhance stability while maintaining critical functions like a constant PID sampling rate. In particular, the relationship between graphics libraries and freeRTOS tasks should be investigated to allow for subsequent processing of user interface actions and heating actions.



Appendix
